

AREZOO RAJABI

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AI/ML Researcher and Engineer

with expertise in AI safety, LLM evaluation, multimodal AI, and trustworthy machine learning. Experienced building production AI systems serving 27,000+ users while designing evaluation frameworks, safety analyses, and scalable experimentation pipelines. PhD in Computer Science with publications at CCS, PETS, and AsiaCCS, bridging frontier AI research with production engineering.

AREAS OF EXPERTISE

LLM Evaluation, AI TrustWorthy, Prompt Engineering, RAG, Agentic AI, Model Fine-Tuning

SKILLS

Programming Languages:	Python, R, SQL, Matlab
Generative AI / LLM:	Prompt engineering (few-shot prompting, chain-of-thought), RAG, NLP (Tokenization, Embeddings), Transformers
AI Safety / Trustworthy AI:	RAG, LSTMs, NLP (Tokenization, Embeddings), PyTorch, Transformers Adversarial ML, Backdoor/Trojan Detection, Membership Inference, Privacy-Preserving ML, Federated Learning
Evaluation & Experimentation:	Benchmarking, Human-in-the-Loop Evaluation, Error Analysis, A/B Testing, KPI metrics Design, Model Monitoring
MLOps & Automation:	MLflow (experiment tracking, model registry), Docker, Git/GitHub, CI/CD pipelines (Github Actions)
Cloud & Data Engineering Tools:	GCP, H2O, Hadoop
Machine Learning Concepts:	Statistical Modeling, Experimental Design, Visualization, Pandas, PySpark, Decision trees, SVM, Regression, Clustering, Gradient Boosting

EXPERIENCE

Senior AI/ML Engineer Wells Fargo	Dec 2022 - Present <i>SF, CA, USA</i>
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- Interactive Chatbot - an enterprise LLM chatbot serving 27,000+ users with \$6M in ROI
 - Built automated LLM evaluation pipelines and curated benchmark datasets to measure response quality, relevance, factuality, and hallucination rates across production workloads.
 - Designed systematic prompt optimization and experimentation frameworks that improved response quality and reduced hallucinations across enterprise LLM workflows.
 - Implemented offline and human-in-the-loop evaluation frameworks to systematically identify model weaknesses and improve performance.
 - Designed online and offline quality metrics and monitoring frameworks to continuously track model performance and detect quality degradation.
- Video Analysis for Suspicious Behavior & Policy Violation Detection
 - Led development of a production multimodal AI system using Vision-Language Models for enterprise safety and policy-violation detection.
 - Designed scalable video data collection, annotation, segmentation, and preprocessing pipelines for large-scale multimodal model evaluation and deployment.
 - Reduced multimodal inference/token costs by 80% through intelligent frame selection and automated video segmentation while preserving detection performance.
 - Developed context-aware metadata and prompting strategies to improve model grounding, detection precision, and robustness.

- Applied ML Solutions:
 - Built predictive models for fraud detection, anomaly detection, next day demand forecasting
 - Performed feature engineering, model evaluation, error analysis, and threshold optimization to reduce false alarms and improve operational efficiency

Postdoctoral Scholar

NSL Lab, University of Washington

Feb 2021 - Dec 2022

Seattle, WA, USA

- Multi-Domain Trojan Detection: Proposed a multi-domain Trojan detection framework across text, image, and audio models, achieving 85% detection accuracy through extensive benchmarking under adversarial settings.
- Privacy-Preserving RL Algorithm: developed and evaluated privacy-preserving reinforcement learning algorithms, conducting extensive experiments to quantify privacy-utility tradeoffs under adversarial settings.
- Federated Learning for Trojan Prevention: Proposed a federated learning approach to counteract Trojan samples during the training phase.

EDUCATION

Ph.D. in Computer Science, Oregon State University 2014 - 2021

Master of Computer Engineering, Sharif University of Technology 2011 - 2013

Bachelor of Computer Science, Sharif University of Technology 2005 - 2010

SELECTED PUBLICATION

1. **A. Rajabi**, S. Asokraj, F. Jiang, L. Niu, B. Ramasubramanian, J. Ritcey, R. Poovendran, MDTD: A Multi-Domain Trojan Detector for Deep Neural Networks, ACM Conference on Computer and Communications Security (ACM CCS), Sep. 2023.
2. J. Jia, Z. Yuan, D. Sahabandu, L. Niu, **A. Rajabi**, B. Ramasubramanian, B. Li, R. Poovendran, FLGAME: A Game-theoretic Defense against Backdoor Attacks In Federated Learning, Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS), Sep 2023 (<https://neurips.cc/virtual/2023/poster/70499>).
3. **A. Rajabi**, D. Sahabandu, L. Niu, B. Ramasubramanian, R. Poovendran, LDL: A Defense for Label-Based Membership Inference Attacks, ACM Asia Conference on Computer and Communications Security (AsiaCCS), July 2023 (7% acceptance rate).
4. **A. Rajabi**, B. Ramasubramanian, A. Marruf, R. Poovendran, Privacy Preserving Reinforcement Learning Beyond Expectation, Accepted in 61st IEEE Conference on Decision and Control, 2022. (<https://arxiv.org/pdf/2203.10165.pdf>).
5. **A. Rajabi**, M. Abbasi, R. B. Bobba, K. Tajik, Adversarial Images Against Super-Resolution Convolutional Neural Networks for Free, Privacy Enhancing Technology Symposium (PETS), 2022.
6. M. Abbasi, **A. Rajabi**, C. Shui, C. Gagné, R. B. Bobba, Toward Adversarial Robustness by Diversity in an Ensemble of Specialized Deep Neural Networks, Canadian Conference on Artificial Intelligence (Canadian AI), 2020. (Best Paper Award)

PATENT

Arezo Rajabi, Dinuka Sahabandu, Luyao Niu, Bhaskar Ramasubramanian, Radha Poovendran, *LDL: A Defense for Label-Based Membership Inference Attacks*, Record of Innovation filed with CoMotion At University of Washington, Seattle Dec. 2022.

PROFESSIONAL SERVICES

- Adjunct Faculty at University of Washington 2022- 2024
- Organizer at The Trojan Detection Challenge (LLM Edition), [NeurIPS 2023](#) 2023
- Organizer at Trojan Detection Challenge , [NeurIPS 2022](#) 2022